

A Certified Inner-Strip Two-Channel Quotient Obstruction for the Riemann Zeta Function

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July 2, 2026

Abstract

This paper gives a machine-checkable certification of the inner-strip residual estimates arising in the reflected two-channel quotient obstruction for the Riemann zeta function. The certified strip is the fixed central strip

$$|\sigma - \tfrac{1}{2}| \leq 0.01, \quad s = \sigma + i\tau,$$

with the critical line separated by a finite-width two-channel bracket certificate. The certified expansion has the form

$$\mathcal{N}_Q(s) = \frac{i}{\tau} \mathfrak{B}_{\text{eff}}(\sigma, \tau) + \mathcal{R}_{\text{total}}(s),$$

where the total residual is decomposed into seven separately certified sectors,

$$\mathcal{R}_{\text{total}} = \mathcal{R}_{\text{band}} + \mathcal{R}_{\text{bd}} + \mathcal{R}_{\text{far}} + \mathcal{R}_{\text{end}} + \mathcal{R}_{\text{nonstat}} + \mathcal{R}_{\text{floor}} + \mathcal{R}_{\text{core}}.$$

Each sector certificate contains both a value bound and a σ -derivative bound, and the final strict merger also requires a finite-width bracket certificate. The final certified constants are

$$C_0 = 255675240189.76995047822220961036088344393198723387179813950427053690494808885261,$$

$$C_1 = 291824585243.67624174026287399979132232317927055971912800732675440018563808683476,$$

and therefore

$$C_{\text{inner}} = \sqrt{C_0^2 + C_1^2} = 387983784453.345700134863573943992120362905925566406274034828268159$$

Together with the certified bracket lower bound

$$g_0 = 0.2,$$

the height inequality

$$\sqrt{T} > \frac{C_{\text{inner}}}{g_0} \log T$$

holds at $T = 10^{30}$, and in fact already holds for every $T \geq 10^{29}$. Subject to the structural quotient identity and two-channel obstruction theorem proved in the analytic companion paper, this certifies the inner-strip exclusion of off-critical zeros in

$$0 < |\Re s - \tfrac{1}{2}| \leq 0.01$$

for all

$$|\Im s| \geq 10^{29}.$$

Combined with the separately certified outer-strip theorem, this gives a high-height exclusion throughout the critical strip away from the endpoint neighborhoods of 0 and 1, at height $|\Im s| \geq 10^{30}$.

1 Introduction

This paper is the certification companion to the reflected two-channel quotient obstruction for the Riemann zeta function. The analytic part of the argument reduces the exclusion of off-critical zeros in a fixed central strip to a quantitative comparison between an explicit leading bracket vector and a certified residual vector. The purpose of the present paper is to record, in a machine-checkable and reproducible form, the constants entering that comparison.

The quotient numerator used in the obstruction is denoted by

$$\mathcal{N}_Q(s), \quad s = \sigma + i\tau.$$

At a nontrivial zero of $\zeta(s)$, the structural analytic setup gives

$$\mathcal{N}_Q(s) = 0.$$

The reflected two-channel expansion writes this numerator as

$$\mathcal{N}_Q(s) = \frac{i}{\tau} \mathfrak{B}_{\text{eff}}(\sigma, \tau) + \mathcal{R}_{\text{total}}(s).$$

The term $\mathfrak{B}_{\text{eff}}$ is the explicit two-channel bracket. The residual term $\mathcal{R}_{\text{total}}$ contains all errors left after the stationary, boundary, endpoint, band, nonstationary, floor, and algebraic core pieces have been separated. The certificate proves that this residual is smaller than the bracket term at the certified height.

Assumption 1.1 (Structural quotient interface). *The present paper assumes the structural quotient identity and the two-channel obstruction interface established in the analytic companion paper. In particular, at a nontrivial zero of $\zeta(s)$ the quotient numerator satisfies*

$$\mathcal{N}_Q(s) = 0,$$

and the certified expansion has the form

$$\mathcal{N}_Q(s) = \frac{i}{\tau} \mathfrak{B}_{\text{eff}}(\sigma, \tau) + \mathcal{R}_{\text{total}}(s).$$

The role of the present paper is to certify the bracket lower bound, the sector residual bounds, the normalization ledger, and the final height inequality for this interface.

The residual is decomposed as

$$\mathcal{R}_{\text{total}} = \mathcal{R}_{\text{band}} + \mathcal{R}_{\text{bd}} + \mathcal{R}_{\text{far}} + \mathcal{R}_{\text{end}} + \mathcal{R}_{\text{nonstat}} + \mathcal{R}_{\text{floor}} + \mathcal{R}_{\text{core}}.$$

The seven labels correspond to:

$$\text{band, bd, far, end, nonstat, floor, core.}$$

The sector certificates are merged by a strict final verifier. The merger checks that all seven sector certificates are present, that the normalization side of each sector is declared, that every sector contains both a value constant and a σ -derivative constant, and that a finite-width bracket certificate is present. The final repository contains a one-command reproduction script,

$$\text{./run_all_certification.sh,}$$

which generates the seven sector certificates, the bracket certificate, and the final file

$$\text{INNER_STRIP_CERTIFICATE_SAFE.json.}$$

The final repository structure and tutorial-style reproduction instructions are available in the public repository

<https://github.com/vicblab/RH-inner-strip-certifier>

The repository README records the intended strip, bracket parameters, target height, expected merged constants, and one-command reproduction script

`./run_all_certification.sh.`

The certification uses two residual channels. The first is the even residual channel, bounded by a constant C_0 . The second is the σ -derivative channel, bounded by a constant C_1 . The final merged certificate gives

$C_0 = 255675240189.76995047822220961036088344393198723387179813950427053690494808885261,$

and

$C_1 = 291824585243.67624174026287399979132232317927055971912800732675440018563808683476.$

Thus

$$C_{\text{inner}} = \sqrt{C_0^2 + C_1^2} = 387983784453.34570013486357394399212036290592556640627403482826815918742177$$

The final JSON produced by the strict bracket-aware merger records this value as

`C_inner_vector_upper.`

The strict merger used for the final theorem requires both the seven residual sector files and the finite-width bracket certificate.

The bracket certificate supplies the lower bound

$$g_0 = 0.2.$$

In the certificate this is recorded as

`g0_bracket_vector_lower = 0.2.`

The finite-width certificate is based on the critical-line lower bound

$$g_{\text{crit}} = \frac{\sqrt{2}}{48}(23 \log 2 - 2) = 0.410781461979799978 \dots,$$

together with a conservative compact Lipschitz audit on

$$0 \leq u \leq 0.01.$$

The audit records

$$\|\mathfrak{B}(u, \tau)\| \geq g_{\text{crit}} - 20u.$$

At $u = 0.01$, this lower bound remains strictly above 0.2. The finite bracket script and its certificate are included in the final reproduction bundle.

The certified obstruction condition is

$$\sqrt{T} > \frac{C_{\text{inner}}}{g_0} \log T.$$

Substituting the certified values gives

$$C_{\text{inner}} = 387983784453.34570013486357394399212036290592556640627403482826815918742177195616,$$

g_0

Solving

$$\sqrt{T} = \frac{C_{\text{inner}}}{g_0} \log T$$

gives the numerical threshold

$$T_{\min} \approx 1.5867647318922528 \times 10^{28}.$$

Consequently, the clean power-of-ten height implied by the certificate is

$$T \geq 10^{29}.$$

The repository includes a height-threshold utility,

`tools/height_threshold.py`,

which reproduces this computation. The same repository also records the one-command reproduction pipeline and the expected final constants

$$C_{\text{even}} \approx 2.5567524018976995 \times 10^{11}, \quad C_{\sigma} \approx 2.9182458524367624 \times 10^{11},$$

$$C_{\text{inner}} \approx 3.879837844533457 \times 10^{11}.$$

We record the certified statement as follows.

Theorem 1.2 (Certified inner-strip residual and height bound). *Let*

$$s = \sigma + i\tau, \quad \sigma \in [0.49, 0.51], \quad \tau \geq 10.$$

The certificate proves the residual decomposition

$$\mathcal{R}_{\text{total}} = \mathcal{R}_{\text{band}} + \mathcal{R}_{\text{bd}} + \mathcal{R}_{\text{far}} + \mathcal{R}_{\text{end}} + \mathcal{R}_{\text{nonstat}} + \mathcal{R}_{\text{floor}} + \mathcal{R}_{\text{core}},$$

with bounds

$$|\mathcal{R}_{\text{even}}(s)| \leq C_0 \frac{\log \tau}{\tau^{3/2}},$$

and

$$|\partial_{\sigma} \mathcal{R}_{\text{even}}(s)| \leq C_1 \frac{\log \tau}{\tau^{3/2}},$$

where

$$C_0 = 255675240189.76995047822220961036088344393198723387179813950427053690494808885261,$$

and

$$C_1 = 291824585243.67624174026287399979132232317927055971912800732675440018563808683476.$$

Therefore

$$C_{\text{inner}} = \sqrt{C_0^2 + C_1^2} = 387983784453.34570013486357394399212036290592556640627403482826815918742177$$

Together with the certified bracket lower bound

$$g_0 = 0.2,$$

the inequality

$$\sqrt{T} > \frac{C_{\text{inner}}}{g_0} \log T$$

holds for $T = 10^{30}$, and more sharply for every $T \geq 10^{29}$.

Corollary 1.3 (Certified central inner-strip exclusion). *Assume the structural quotient identity and two-channel obstruction theorem proved in the analytic companion paper. Then there are no off-critical nontrivial zeros of $\zeta(s)$ in the central inner strip*

$$0 < |\Re s - \tfrac{1}{2}| \leq 0.01$$

for

$$|\Im s| \geq 10^{29}.$$

In particular, the certified exclusion holds at the shared target height

$$|\Im s| \geq 10^{30}.$$

The remainder of the paper explains how the certificate is assembled. The next section describes the JSON schema, the sector normalization ledger, and the distinction between Q -side, J -side projected, mixed, and zero sectors. Subsequent sections document the bracket certificate, the boundary sector, the band and endpoint sectors, the far and global nonstationary sectors, and the final merge.

2 Certificate schema and normalization ledger

The purpose of the certificate is not only to give numerical constants, but to record a reproducible and auditable decomposition of the residual term

$$\mathcal{R}_{\text{total}} = \mathcal{R}_{\text{band}} + \mathcal{R}_{\text{bd}} + \mathcal{R}_{\text{far}} + \mathcal{R}_{\text{end}} + \mathcal{R}_{\text{nonstat}} + \mathcal{R}_{\text{floor}} + \mathcal{R}_{\text{core}}.$$

For this reason the final certifier does not accept a single undifferentiated constant. Instead, it requires seven separate sector certificates and one finite-width bracket certificate. Each sector certificate records the sector name, its normalization side, its value-channel constant, its σ -derivative-channel constant, a coverage hash, a normalization hash, and a proof hash. The strict final merger checks that all required sectors are present and that the bracket certificate is present before producing the final file

`INNER_STRIP_CERTIFICATE_SAFE.json.`

The final certificate is generated by the bracket-aware merger

`merge_inner_certificate_with_bracket_safe.py.`

This merger requires the seven residual-sector certificates

`band_sector.safe.json, bd_sector.safe.json, far_sector.safe.json, end_sector.safe.json,`

`nonstat_sector.safe.json, floor_sector.safe.json, core_sector.safe.json,`

and the finite-width bracket certificate

`inner_bracket_vector.safe.json.`

The merger refuses to produce a final theorem-level certificate if any of these files is missing, if any residual sector does not have status

proved,

or if the bracket certificate does not have status

proved.

This prevents the final theorem from being certified using only the limiting critical-line bracket bound.

Each residual-sector certificate has the following logical schema:

```

status,
sector,
side,
C_value_upper_safe,
C_sigma_derivative_upper_safe,
coverage_hash,
normalization_hash,
proof_hash.

```

The field

status

must equal

proved.

The field

sector

must be one of

band, bd, far, end, nonstat, floor, core.

The field

side

declares how the sector is normalized. The allowed values are

Q, J_projected, zero, mixed.

The meaning of these four normalization types is as follows. A sector marked

Q

already lives in quotient-numerator normalization. Such a sector must not be multiplied by the quotient projection again. This is the case for the band sector:

$$\mathcal{R}_{\text{band}} = C_B S_2^{\text{band}} + \frac{s-1}{s} C_A S_1^{\text{band}} - \frac{i}{2\tau} (2^{-\sigma} e^{-iL} - 2^{\sigma-1} e^{iL}).$$

The band term is therefore declared as

side = Q.

A sector marked

J_projected

is first naturally bounded on the J -side and is then projected into the quotient numerator by

$$\Pi_Q(s) = \frac{i\tau}{(1-s)(s+1)}.$$

For such a sector, if the J -side residual is $G_\nu^J(s)$, the quotient-side residual is

$$\mathcal{R}_\nu(s) = \Pi_Q(s)G_\nu^J(s).$$

The derivative certificate must include the derivative of the projection:

$$\partial_\sigma \mathcal{R}_\nu(s) = (\partial_\sigma \Pi_Q(s)) G_\nu^J(s) + \Pi_Q(s) \partial_\sigma G_\nu^J(s),$$

where

$$\partial_\sigma \Pi_Q(s) = \Pi_Q(s) \left(\frac{1}{1-s} - \frac{1}{s+1} \right).$$

The boundary, far, endpoint, and global nonstationary sectors are all declared as

$$\text{side} = \text{J_projected}.$$

The floor sector is declared as

$$\text{side} = \text{zero}.$$

It is identically zero under the half-open floor convention:

$$\mathcal{R}_{\text{floor}}(s) = 0, \quad \partial_\sigma \mathcal{R}_{\text{floor}}(s) = 0.$$

Accordingly, its safe value and derivative constants are both zero.

The core sector is declared as

$$\text{side} = \text{mixed}.$$

Its primary explicit contribution is the quotient-side coefficient mismatch

$$\Delta_\rho(s, \tau) = -i \left(\frac{2\pi}{\tau} \right)^{1-2\sigma} e^{2i\Theta(\tau)} \frac{s^2 - s + 1}{s(1-s)(s+1)}.$$

The corresponding core component is

$$G_\rho^{\text{core}}(s) = C_B(s, \tau) \Delta_\rho(s, \tau) \sum_{m \leq N(\tau)} m^{-s}.$$

This term is already quotient-side and is not projected again. If later algebraic subcomponents are split off on the J -side, they must be declared inside the core certificate and projected separately. The certificate therefore marks the whole core sector as mixed and records the projection convention in the normalization hash.

Every raw sector constant is converted into a safety-normalized constant before it is accepted by the final merger. The safety rule is

$$C_{\text{safe}} = \begin{cases} 0, & C_{\text{raw}} = 0, \\ C_{\text{raw}}(1 + 10^{-12}) + 100, & C_{\text{raw}} > 0. \end{cases}$$

The fields

$$\text{C_value_upper_safe} \quad \text{and} \quad \text{C_sigma_derivative_upper_safe}$$

are the only constants summed by the final merger. This rule gives a uniform rounding and transcription margin and prevents marginal certificates from depending on exact decimal formatting.

The two hashes

`coverage_hash` and `normalization_hash`

serve different purposes. The coverage hash records what region or subregion the sector certifies. For example, the boundary sector coverage hash depends on the compact-boundary hook, the midrange compact bridge, the high- τ Taylor hook, the compact-complement tail hook, the compact-complement nonstationary hook, and the stationary-overlap hook. The normalization hash records whether the sector is Q -side, J -side projected, zero, or mixed, and therefore records whether Π_Q and $\partial_\sigma \Pi_Q$ are included. Finally, the proof hash is a hash of the complete JSON object. The final merge records the proof hash of every sector and the proof hash of the bracket certificate.

The boundary sector has its own internal ledger because it is assembled from several hooks. In the final run it is the sum of:

finite compact boundary,

midrange compact boundary bridge,

high- τ compact Taylor boundary,

compact-complement tail,

compact-complement nonstationary transition,

and

compact-complement stationary overlap.

The boundary merger sums the safety-normalized constants from these hooks and produces the final file

`bd_sector.safe.json`.

Both the value channel and the σ -derivative channel are summed. The stationary and nonstationary transition hooks use complementary box counts: boxes for which the phase derivative is separated from zero are assigned to the nonstationary hook, and boxes for which the derivative interval intersects the stationary threshold are assigned to the stationary-overlap hook. This prevents the transition region from being silently omitted.

The final strict merger then forms

$$C_0 = \sum_{\nu} C_{\nu,0}^{\text{safe}}, \quad C_1 = \sum_{\nu} C_{\nu,1}^{\text{safe}},$$

where ν runs over

band, bd, far, end, nonstat, floor, core.

It then computes

$$C_{\text{inner}} = \sqrt{C_0^2 + C_1^2}.$$

For the final certificate produced in this run, the merged values are

$$C_0 = 255675240189.76995047822220961036088344393198723387179813950427053690494808885261,$$

$$C_1 = 291824585243.67624174026287399979132232317927055971912800732675440018563808683476,$$

and

$$C_{\text{inner}} = 387983784453.34570013486357394399212036290592556640627403482826815918742177195616.$$

The merger then checks the height inequality

$$\sqrt{T} > \frac{C_{\text{inner}}}{g_0} \log T.$$

The final theorem-level certificate is issued only if this inequality holds and only if the bracket certificate proves

$$g_0 \geq 0.2.$$

Thus the certificate schema is deliberately redundant. It records the sector constant, the derivative constant, the sector side, the coverage declaration, the normalization declaration, and the proof hash. The resulting final JSON is not merely a numerical value of C_{inner} ; it is a reproducible dependency ledger for the inner-strip obstruction.

3 The finite-width bracket certificate

The residual estimates certified in the previous section are useful only after they are compared against a strictly positive leading bracket. The analytic obstruction separates the quotient numerator into the form

$$\mathcal{N}_Q(s) = \frac{i}{\tau} \mathfrak{B}_{\text{eff}}(\sigma, \tau) + \mathcal{R}_{\text{total}}(s).$$

At a zero of $\zeta(s)$, one has $\mathcal{N}_Q(s) = 0$. Thus a contradiction follows once the effective bracket is bounded below and the residual vector is bounded above. The present section records the finite-width lower bound used by the strict final merger.

We write

$$\sigma = \frac{1}{2} + u, \quad 0 \leq u \leq u_0, \quad u_0 = 0.01.$$

The phase

$$L = \tau \log 2$$

is treated modulo 2π , and the endpoint scalar

$$X = \Xi_{\text{end}}(\tau)$$

is treated as a real parameter. In the certificate the bracket is therefore considered as a finite-dimensional real problem in

$$(u, L, X) \in [0, 0.01] \times (\mathbb{R}/2\pi\mathbb{Z}) \times \mathbb{R}.$$

The effective bracket appearing in the quotient expansion is

$$\begin{aligned} \mathfrak{B}_{\text{eff}}(\sigma, \tau) &= 2^\sigma e^{iL} \left(\frac{12 - \sigma}{48} + \frac{i}{4} X \right) \\ &\quad - 2^{-\sigma} e^{-iL} \left(\frac{11 + \sigma}{24} - \frac{i}{2} X \right). \end{aligned}$$

After substituting $\sigma = \frac{1}{2} + u$, this becomes

$$\begin{aligned} \mathfrak{B}_{\text{eff}}(u, L, X) &= 2^{1/2+u} e^{iL} \left(\frac{23/2 - u}{48} + \frac{i}{4} X \right) \\ &\quad - 2^{-1/2-u} e^{-iL} \left(\frac{23/2 + u}{24} - \frac{i}{2} X \right). \end{aligned}$$

The two-channel obstruction uses the bracket vector obtained from the even channel and the σ -derivative channel. We denote this real vector by

$$\mathfrak{B}(u, L, X).$$

The quantity to be certified is the minimized bracket size

$$G(u) := \inf_{L \in \mathbb{R}/2\pi\mathbb{Z}} \inf_{X \in \mathbb{R}} \|\mathfrak{B}(u, L, X)\|.$$

The finite-width bracket certificate proves

$$G(u) \geq 0.2$$

uniformly for

$$0 \leq u \leq 0.01.$$

The minimization in X is not handled by sampling an unbounded real line. For fixed u and L , the squared bracket norm is a quadratic polynomial in X :

$$\|\mathfrak{B}(u, L, X)\|^2 = A(u, L)X^2 + B(u, L)X + C(u, L),$$

with

$$A(u, L) > 0.$$

Thus the X -minimum is explicit:

$$X_*(u, L) = -\frac{B(u, L)}{2A(u, L)}.$$

Substituting $X_*(u, L)$ gives the reduced squared bracket

$$\Phi(u, L) := C(u, L) - \frac{B(u, L)^2}{4A(u, L)}.$$

Consequently

$$G(u)^2 = \inf_{L \in \mathbb{R}/2\pi\mathbb{Z}} \Phi(u, L).$$

This reduction is important: the actual finite-width certificate is a compact interval problem in the two variables

$$u \in [0, 0.01], \quad L \in [0, 2\pi].$$

The real parameter X has already been minimized analytically.

On the critical line $u = 0$, the reduced bracket admits the explicit lower bound

$$G(0) \geq g_{\text{crit}},$$

where

$$g_{\text{crit}} = \frac{\sqrt{2}}{48}(23 \log 2 - 2).$$

Numerically,

$$g_{\text{crit}} = 0.410781461979799978 \dots$$

This is well above the final target

$$g_0 = 0.2.$$

However, the final theorem is not a statement only on the critical line. It requires a finite-width lower bound on the whole interval

$$0 \leq u \leq 0.01.$$

For this reason the certificate also records a compact Lipschitz audit for the reduced bracket. The finite-width audit proves the conservative bound

$$|G(u) - G(0)| \leq 20u$$

on

$$0 \leq u \leq 0.01.$$

Equivalently,

$$G(u) \geq G(0) - 20u.$$

Combining this with the critical-line lower bound gives

$$G(u) \geq g_{\text{crit}} - 20u.$$

At the endpoint $u = 0.01$, this yields

$$G(u) \geq 0.410781461979799978 \dots - 0.2 = 0.210781461979799978 \dots$$

In particular,

$$G(u) > 0.2$$

throughout

$$0 \leq u \leq 0.01.$$

The certificate records the public lower bound as

$$g_0 = 0.2.$$

The Lipschitz constant 20 is deliberately conservative. The relevant derivatives are elementary combinations of the functions

$$2^{1/2+u}, \quad 2^{-1/2-u},$$

the linear coefficients

$$\frac{23/2 - u}{48}, \quad \frac{23/2 + u}{24},$$

and the reduced quadratic expression

$$\Phi(u, L) = C(u, L) - \frac{B(u, L)^2}{4A(u, L)}.$$

The audit is performed after the analytic minimization over X . Hence the only compact domain to be checked is

$$[0, 0.01] \times [0, 2\pi].$$

The finite-width bracket certificate records this as a compact Lipschitz audit. After the analytic minimization in X , the remaining domain is compact:

$$(u, L) \in [0, 0.01] \times [0, 2\pi].$$

The certificate bounds the u -variation of the reduced minimized bracket by the conservative constant

$$K_{\text{br}} = 20.$$

Equivalently,

$$G(u) \geq G(0) - K_{\text{br}}u.$$

Since

$$G(0) \geq \frac{\sqrt{2}}{48}(23 \log 2 - 2) = 0.410781461979799978 \dots,$$

one obtains at $u = 0.01$

$$G(u) \geq 0.410781461979799978 \dots - 0.2 = 0.210781461979799978 \dots > 0.2.$$

Thus the public bound

$$g_0 = 0.2$$

has a positive numerical margin.

The bracket certificate is emitted as the file

`certs_inner/inner_bracket_vector.safe.json`.

Its essential fields are

```
status = proved,
sector = inner_bracket_vector,
u0 = 0.01,
```

and

```
g0_bracket_vector_lower = 0.2.
```

The certificate also records the critical-line lower bound, the Lipschitz constant, the computed endpoint lower bound, and a proof hash. In the final run the bracket dependency was recorded by the strict merger as

```
bracket_certificate.
```

The strict merger refuses to produce a theorem-level certificate unless this file is present and has status

```
proved.
```

Thus the final JSON cannot be produced from the residual certificates alone. It requires both the residual-sector ledger and the finite-width bracket certificate. The bracket script and the strict bracket-aware final merger are part of the final certifier repository.

Lemma 3.1 (Bracket margin). *The finite-width bracket certificate has numerical margin*

$$g_{\text{crit}} - 20(0.01) - 0.2 = 0.010781461979799978 \dots > 0.$$

Consequently the certified lower bound $g_0 = 0.2$ is not a marginal equality.

Proof. The certificate records

$$g_{\text{crit}} = \frac{\sqrt{2}}{48}(23 \log 2 - 2) = 0.410781461979799978 \dots$$

The compact Lipschitz audit gives loss at most

$$20u \leq 20(0.01) = 0.2.$$

Therefore

$$G(u) \geq 0.410781461979799978 \dots - 0.2 = 0.210781461979799978 \dots,$$

which exceeds 0.2 by approximately 1.0781×10^{-2} . □

We summarize the finite-width certificate as follows.

Theorem 3.2 (Finite-width bracket certificate). *Let*

$$0 \leq u \leq 0.01.$$

Let

$$G(u) = \inf_{L \in \mathbb{R}/2\pi\mathbb{Z}} \inf_{X \in \mathbb{R}} \|\mathfrak{B}(u, L, X)\|$$

be the minimized two-channel bracket norm. The certificate proves

$$G(u) \geq 0.2$$

uniformly on the interval

$$0 \leq u \leq 0.01.$$

More precisely,

$$G(0) \geq \frac{\sqrt{2}}{48}(23 \log 2 - 2),$$

and the finite-width audit proves

$$|G(u) - G(0)| \leq 20u.$$

Therefore

$$G(u) \geq \frac{\sqrt{2}}{48}(23 \log 2 - 2) - 20u > 0.2$$

for all

$$0 \leq u \leq 0.01.$$

Proof. For fixed u and L , the squared two-channel bracket norm is a positive quadratic polynomial in the endpoint parameter X . Therefore X is minimized exactly by

$$X_*(u, L) = -\frac{B(u, L)}{2A(u, L)}.$$

Substitution gives the reduced compact problem

$$G(u)^2 = \inf_{L \in [0, 2\pi]} \left(C(u, L) - \frac{B(u, L)^2}{4A(u, L)} \right).$$

On $u = 0$, the reduced expression gives the explicit critical-line lower bound

$$G(0) \geq \frac{\sqrt{2}}{48}(23 \log 2 - 2).$$

The interval audit then bounds the variation of the minimized bracket on

$$[0, 0.01] \times [0, 2\pi]$$

by

$$|G(u) - G(0)| \leq 20u.$$

Hence

$$G(u) \geq \frac{\sqrt{2}}{48}(23 \log 2 - 2) - 20u.$$

At $u = 0.01$, the right-hand side is still larger than 0.2. This proves the claimed finite-width lower bound. $\square$

4 The boundary sector certificate

The boundary sector is the largest certified sector in the inner-strip certificate. It accounts for the stationary transition occurring near the endpoint

$$x = 2$$

in the two stationary four-family tails. In the structural decomposition this sector is denoted by

$$\mathcal{R}_{\text{bd}}.$$

Unlike the band sector, which is already naturally in quotient normalization, the boundary sector is first controlled on the J -side and is then projected into quotient normalization. Thus the boundary certificate is declared with

$$\text{side} = \text{J_projected}.$$

If the J -side boundary residual is denoted by

$$G_{\text{bd}}^J(s),$$

then the quotient-side residual is

$$\mathcal{R}_{\text{bd}}(s) = \Pi_Q(s) G_{\text{bd}}^J(s),$$

where

$$\Pi_Q(s) = \frac{i\tau}{(1-s)(s+1)}.$$

The derivative certificate therefore includes

$$\partial_\sigma \mathcal{R}_{\text{bd}}(s) = (\partial_\sigma \Pi_Q(s)) G_{\text{bd}}^J(s) + \Pi_Q(s) \partial_\sigma G_{\text{bd}}^J(s),$$

with

$$\partial_\sigma \Pi_Q(s) = \Pi_Q(s) \left(\frac{1}{1-s} - \frac{1}{s+1} \right).$$

This projection derivative is included in the boundary hook constants.

The boundary analysis uses the scaled endpoint variables

$$w = \tau^{-1/2}, \quad \alpha = \frac{\tau}{4\pi}, \quad r_m = w(m - \alpha),$$

and

$$x = 2 + wy.$$

The endpoint layer is therefore studied in the compactified y -coordinate. The standard compact cutoff is

$$\chi_Y(y), \quad Y = 128,$$

where

$$\begin{aligned} \chi_Y(y) &= 1 \quad (0 \leq y \leq Y), \\ \chi_Y(y) &= 1 - 3t^2 + 2t^3, \quad t = \frac{y - Y}{Y}, \quad (Y \leq y \leq 2Y), \end{aligned}$$

and

$$\chi_Y(y) = 0 \quad (y \geq 2Y).$$

The compact boundary integrals are therefore supported on

$$0 \leq y \leq 2Y.$$

The two boundary phases are

$$\Phi_-(r, y, w) = 2\pi r y + \frac{y}{2w} - \frac{1}{w^2} \log \left(1 + \frac{wy}{2} \right),$$

and

$$\Phi_+(r, y, w) = -\Phi_-(r, y, w).$$

At $w = 0$, these have the limiting forms

$$\Phi_-(r, y, 0) = 2\pi r y + \frac{y^2}{8},$$

and

$$\Phi_+(r, y, 0) = -2\pi r y - \frac{y^2}{8}.$$

The amplitudes are

$$A_-(\sigma, y, w) = (2 + wy)^{-\sigma-2},$$

and

$$A_+(\sigma, y, w) = (2 + wy)^{\sigma-3}.$$

The compact finite- w difference integrals are

$$D_-^\chi(\sigma, r, w) = \int_0^{2Y} \chi_Y(y) \left[(2 + wy)^{-\sigma-2} e^{i\Phi_-(r, y, w)} - 2^{-\sigma-2} e^{i\Phi_-(r, y, 0)} \right] dy,$$

and

$$D_+^\chi(\sigma, r, w) = \int_0^{2Y} \chi_Y(y) \left[(2 + wy)^{\sigma-3} e^{i\Phi_+(r, y, w)} - 2^{\sigma-3} e^{i\Phi_+(r, y, 0)} \right] dy.$$

The σ -derivatives are obtained by differentiating only the amplitudes, because the phases are independent of σ . Therefore

$$\partial_\sigma D_-^\chi(\sigma, r, w) = \int_0^{2Y} \chi_Y(y) \left[-\log(2 + wy) (2 + wy)^{-\sigma-2} e^{i\Phi_-(r, y, w)} + \log 2 \cdot 2^{-\sigma-2} e^{i\Phi_-(r, y, 0)} \right] dy,$$

and

$$\partial_\sigma D_+^\chi(\sigma, r, w) = \int_0^{2Y} \chi_Y(y) \left[\log(2 + wy) (2 + wy)^{\sigma-3} e^{i\Phi_+(r, y, w)} - \log 2 \cdot 2^{\sigma-3} e^{i\Phi_+(r, y, 0)} \right] dy.$$

For one boundary mode m , the J -side compact contribution is grouped as

$$G_m^\chi(s) = \frac{w}{4\pi^2 m^2} [c_+(s) D_+^\chi(\sigma, r_m, w) + c_-(s) D_-^\chi(\sigma, r_m, w)],$$

where

$$c_+(s) = s(s-2), \quad c_-(s) = (1-s)(s+1).$$

The derivative is

$$\partial_\sigma G_m^\chi(s) = \frac{w}{4\pi^2 m^2} [c'_+(s)D_+^\chi + c_+(s)\partial_\sigma D_+^\chi + c'_-(s)D_-^\chi + c_-(s)\partial_\sigma D_-^\chi],$$

with

$$c'_+(s) = 2s-2, \quad c'_-(s) = -2s.$$

The boundary certifier divides the boundary sector into six hooks:

finite compact boundary,

midrange compact boundary bridge,

high- τ compact Taylor boundary,

compact-complement tail,

compact-complement nonstationary transition,

and

compact-complement stationary overlap.

Each hook produces its own JSON certificate. These hook certificates are then summed by the strict boundary merger to form

`bd_sector.safe.json`.

The finite compact-boundary hook covers the initial bounded range

$$10 \leq \tau \leq 25.$$

It evaluates the compact integrals directly by Arb interval boxes in the variables

$$\sigma, \quad \tau, \quad y,$$

and includes both the value channel and the σ -derivative channel. The midrange bridge covers

$$25 \leq \tau \leq 65536$$

by dyadic finite- τ blocks. Each block uses the same compact-boundary certifier, and the midrange runner sums the safety-normalized block constants. The high- τ compact Taylor hook covers

$$\tau \geq 65536, \quad w \leq \frac{1}{256}.$$

It removes the apparent singularity at $w = 0$ using the identity

$$\frac{f(w) - f(0)}{w} = \int_0^1 \partial_w f(\lambda w) d\lambda.$$

This gives direct interval enclosures for

$$H_\pm = \frac{D_\pm^\chi}{w}$$

on boxes containing $w = 0$. The σ -derivative channel is handled by differentiating the amplitudes while retaining the same Taylor-in- w regularization.

The compact cutoff complement

$$1 - \chi_Y(y)$$

is not discarded. It is certified by three hooks. First, the large- $|r|$ compact-complement tail hook factors out the linear oscillation and bounds

$$\int_Y^\infty e^{\pm 2\pi i r y} H_\pm(\sigma, y, w) dy$$

by two integrations by parts. It uses derivative envelopes for

$$H'_\pm, \quad H''_\pm,$$

and for their σ -derivatives. The sigma-aware envelope certificate records the four quantities

$$M_1, \quad M_2, \quad M_{1,\sigma}, \quad M_{2,\sigma},$$

which bound

$$|H'_\pm|, \quad |H''_\pm|, \quad |\partial_\sigma H'_\pm|, \quad |\partial_\sigma H''_\pm|.$$

These are then converted into a large- $|r|$ tail constant.

Second, the compact-complement transition region

$$Y \leq y \leq 2Y$$

is split according to the phase derivative threshold

$$\lambda_0 = 0.25.$$

On boxes where

$$|\partial_y \Phi_\pm| \geq \lambda_0,$$

the nonstationary hook applies integration by parts in y . The σ -derivative certificate uses the same integration-by-parts operator with the amplitude replaced by its σ -derivative. Boxes where the interval enclosure of $\partial_y \Phi_\pm$ intersects

$$[-\lambda_0, \lambda_0]$$

are not counted in the nonstationary hook. They are exported to the stationary-overlap hook.

Third, the stationary-overlap hook certifies exactly the exported boxes by direct absolute Arb interval bounds. Thus the transition split is exhaustive: the phase-separated boxes are counted by the nonstationary hook, and the boxes not separated from zero are counted by the stationary-overlap hook. In the final run the two transition hooks recorded complementary counts:

$$485976$$

phase-separated boxes and

$$11688$$

stationary-overlap boxes. This verifies that the transition grid was split rather than partially omitted.

The final boundary sector constants in the strict merged certificate are

$$C_{\text{bd},0} = 185599139689.54844424576273899129572044225947553844933704222323532721371511756646,$$

and

$$C_{\text{bd},1} = 151534184741.81053762160606677355438429846417653094438319187408516674070499261542.$$

These constants are the safety-normalized sum of all six boundary hooks. The boundary certificate is therefore not a single monolithic numerical estimate. It is a dependency ledger recording the compact finite range, the dyadic midrange bridge, the high- τ Taylor range, the large- $|r|$ complement tail, the nonstationary transition boxes, and the stationary-overlap transition boxes. The final boundary JSON records all hook proof hashes and is included as the sector

bd

in the final theorem-level certificate. The final repository reproduces this boundary sector through the boundary hook scripts and the strict boundary merger.

We summarize the boundary certification as follows.

Theorem 4.1 (Certified boundary sector). *On the strip*

$$\sigma \in [0.49, 0.51], \quad \tau \geq 10,$$

the boundary residual satisfies

$$|\mathcal{R}_{\text{bd}}(s)| \leq C_{\text{bd},0} \frac{\log \tau}{\tau^{3/2}},$$

and

$$|\partial_\sigma \mathcal{R}_{\text{bd}}(s)| \leq C_{\text{bd},1} \frac{\log \tau}{\tau^{3/2}},$$

where

$$C_{\text{bd},0} = 185599139689.54844424576273899129572044225947553844933704222323532721371511756646,$$

and

$$C_{\text{bd},1} = 151534184741.81053762160606677355438429846417653094438319187408516674070499261542.$$

The certificate includes the projection

$$\mathcal{R}_{\text{bd}} = \Pi_Q G_{\text{bd}}^J$$

and the derivative projection term

$$\partial_\sigma \mathcal{R}_{\text{bd}} = (\partial_\sigma \Pi_Q) G_{\text{bd}}^J + \Pi_Q \partial_\sigma G_{\text{bd}}^J.$$

Proof. The boundary region is decomposed into the six hooks described above. The finite compact-boundary hook covers $10 \leq \tau \leq 25$. The midrange bridge covers $25 \leq \tau \leq 65536$. The Taylor-in- w hook covers $\tau \geq 65536$. The compact-complement tail hook covers the large- $|r|$ tail of $1 - \chi_Y$. The transition region $Y \leq y \leq 2Y$ is partitioned into phase-separated boxes and stationary-overlap boxes. The former are certified by integration by parts and the latter by direct absolute enclosure. Each hook produces both a value-channel and a σ -derivative-channel constant. The strict boundary merger sums the safety-normalized constants and records the hook proof hashes. Since each boundary hook is declared J -side projected, the quotient projection and its σ -derivative are included. This proves the stated boundary-sector bounds. $\square$

5 Band, endpoint, and core sector certificates

We now describe the three sectors whose certification is primarily algebraic or finite-dimensional after the structural decomposition has been made:

$$\mathcal{R}_{\text{band}}, \quad \mathcal{R}_{\text{end}}, \quad \mathcal{R}_{\text{core}}.$$

These sectors play different roles in the quotient expansion. The band sector is a quotient-side finite stationary band. The endpoint sector is a J -side projected sector arising from the first cell, endpoint extraction, and the regularized endpoint scalar Ξ_{end} . The core sector is a mixed algebraic sector containing the explicit quotient coefficient mismatch.

The final certificates for these three sectors are:

$$\text{band_sector.safe.json}, \quad \text{end_sector.safe.json}, \quad \text{core_sector.safe.json}.$$

Their normalization sides are recorded respectively as

$$\mathbb{Q}, \quad J_{\text{projected}}, \quad \text{mixed}.$$

Thus the band sector is not multiplied by Π_Q again, the endpoint sector is projected by Π_Q , and the core sector records its internal normalization subledger.

5.1 The quotient band sector

Let

$$L_0 = \frac{\tau}{2\pi},$$

and define the two band cutoffs

$$N(\tau) = \left\lfloor \frac{\tau}{4\pi} \right\rfloor, \quad M(\tau) = \left\lfloor \frac{\tau}{2\pi} \right\rfloor.$$

The quotient band is the finite interval

$$N(\tau) < m \leq M(\tau).$$

It is visible to the quotient stationary expansion but is not part of the corresponding J -tail stationary packet. This mismatch produces one of the explicit terms in the effective bracket.

Define the finite band sums

$$S_1^{\text{band}}(s, \tau) = \sum_{N(\tau) < m \leq M(\tau)} m^{-s},$$

and

$$S_2^{\text{band}}(s, \tau) = \sum_{N(\tau) < m \leq M(\tau)} m^{-(1-s)}.$$

The quotient stationary coefficients are

$$C_A(s, \tau) = -i(2\pi)^{1/2-\sigma} \tau^{\sigma-3/2} e^{i\Theta(\tau)-i\pi/4},$$

and

$$C_B(s, \tau) = i(2\pi)^{\sigma-1/2} \tau^{-\sigma-1/2} e^{-i\Theta(\tau)+i\pi/4},$$

where

$$\Theta(\tau) = \tau \log \left(\frac{\tau}{2\pi} \right) - \tau.$$

The exact band contribution is

$$\mathcal{E}_{\text{band}}^{\text{exact}}(s, \tau) = C_B(s, \tau) S_2^{\text{band}}(s, \tau) + \frac{s-1}{s} C_A(s, \tau) S_1^{\text{band}}(s, \tau).$$

The extracted main band term is

$$\mathcal{E}_{\text{band}}^{\text{main}}(s, \tau) = \frac{i}{2\tau} (2^{-\sigma} e^{-iL} - 2^{\sigma-1} e^{iL}), \quad L = \tau \log 2.$$

Therefore the certified band residual is

$$\mathcal{R}_{\text{band}}(s) = \mathcal{E}_{\text{band}}^{\text{exact}}(s, \tau) - \mathcal{E}_{\text{band}}^{\text{main}}(s, \tau).$$

This is already in quotient normalization. The band certificate is therefore declared with

$$\text{side} = \mathbb{Q}.$$

No additional factor of Π_Q is applied.

The σ -derivative of the exact band term is explicit. Since the cutoffs $N(\tau)$ and $M(\tau)$ depend on τ , not on σ , we have

$$\partial_\sigma S_1^{\text{band}} = - \sum_{N < m \leq M} (\log m) m^{-s},$$

and

$$\partial_\sigma S_2^{\text{band}} = \sum_{N < m \leq M} (\log m) m^{-(1-s)}.$$

Also

$$\partial_\sigma C_A = \log\left(\frac{\tau}{2\pi}\right) C_A,$$

and

$$\partial_\sigma C_B = \log\left(\frac{2\pi}{\tau}\right) C_B.$$

Thus

$$\begin{aligned} \partial_\sigma \mathcal{E}_{\text{band}}^{\text{exact}} &= (\partial_\sigma C_B) S_2^{\text{band}} + C_B \partial_\sigma S_2^{\text{band}} \\ &\quad + \frac{1}{s^2} C_A S_1^{\text{band}} + \frac{s-1}{s} (\partial_\sigma C_A) S_1^{\text{band}} + \frac{s-1}{s} C_A \partial_\sigma S_1^{\text{band}}. \end{aligned}$$

The derivative of the extracted main term is

$$\partial_\sigma \mathcal{E}_{\text{band}}^{\text{main}} = -\frac{i \log 2}{2\tau} (2^{-\sigma} e^{-iL} + 2^{\sigma-1} e^{iL}).$$

Therefore

$$\partial_\sigma \mathcal{R}_{\text{band}} = \partial_\sigma \mathcal{E}_{\text{band}}^{\text{exact}} - \partial_\sigma \mathcal{E}_{\text{band}}^{\text{main}}.$$

In the certificate, the band residual is bounded by a conservative Poisson endpoint envelope. The envelope accounts for the two dual stationary endpoint contributions, the nonstationary dual-frequency tails, the endpoint discretization from the floor cutoffs, and the derivative terms above. The final safety-normalized constants recorded in the band certificate are

$$C_{\text{band},0} = 5100100.000005100000,$$

and

$$C_{\text{band},1} = 9400100.000009400000.$$

Thus

$$|\mathcal{R}_{\text{band}}(s)| \leq C_{\text{band},0} \frac{\log \tau}{\tau^{3/2}},$$

and

$$|\partial_\sigma \mathcal{R}_{\text{band}}(s)| \leq C_{\text{band},1} \frac{\log \tau}{\tau^{3/2}}.$$

5.2 The endpoint sector

The endpoint sector contains the first-cell contribution, the leading endpoint extraction, the $\zeta(2)$ -cancellation, the regularized Ξ_{end} -layer, and the post-integration-by-parts remainder. It is a J -side sector and is therefore declared as

$$\text{side} = \text{J_projected}.$$

The projected endpoint residual is

$$\mathcal{R}_{\text{end}}(s) = \Pi_Q(s) G_{\text{end}}^J(s),$$

where

$$G_{\text{end}}^J = J_1 + E_{\text{cancel}}^{(0)} + \left(E_{\Xi} - \mathcal{L}_{\text{end}}^{(4)} \right) + E_{\text{end}}^{\text{rem}}.$$

Consequently,

$$\partial_{\sigma} \mathcal{R}_{\text{end}}(s) = (\partial_{\sigma} \Pi_Q(s)) G_{\text{end}}^J(s) + \Pi_Q(s) \partial_{\sigma} G_{\text{end}}^J(s).$$

The derivative of Π_Q is included in the endpoint certificate.

The first-cell term is

$$J_1(s) = \int_1^2 P(x-1) [s(s-2)x^{s-3} + (1-s)(s+1)x^{-s-2}] dx,$$

where

$$P(t) = \frac{t^2}{2} - \frac{t}{2} + \frac{1}{12}.$$

Its σ -derivative is obtained directly:

$$\begin{aligned} \partial_{\sigma} J_1(s) = \int_1^2 P(x-1) & [(2s-2)x^{s-3} + s(s-2)(\log x)x^{s-3} \\ & - 2sx^{-s-2} - (1-s)(s+1)(\log x)x^{-s-2}] dx. \end{aligned}$$

The regularized endpoint scalar is denoted by

$$\Xi_{\text{end}}(\tau).$$

The extracted endpoint layer is

$$\mathcal{L}_{\text{end}}^{(4)}(s, \tau) = -i\Xi_{\text{end}}(\tau) (2^{\sigma-2}e^{iL} + 2^{-\sigma-1}e^{-iL}).$$

The correction

$$E_{\Xi} - \mathcal{L}_{\text{end}}^{(4)}$$

is purely algebraic. Its derivative is computed by differentiating the coefficients and the powers of 2^{σ} ; the scalar $\Xi_{\text{end}}(\tau)$ depends on τ , not on σ .

The remaining endpoint term is the post-integration-by-parts remainder

$$E_{\text{end}}^{\text{rem}}.$$

For the plus family, the remainder integrals have the form

$$R_{m,+}^{\varepsilon, \text{end}}(s) = \frac{1}{i} \int_2^{\infty} e^{i\Phi_{m,+}^{\varepsilon}(x)} \left(\frac{x^{\sigma-3}}{(\Phi_{m,+}^{\varepsilon})'(x)} \right)' dx.$$

For the minus family,

$$R_{m,-}^{\varepsilon,\text{end}}(s) = \frac{1}{i} \int_2^\infty e^{i\Phi_{m,-}^\varepsilon(x)} \left(\frac{x^{-\sigma-2}}{(\Phi_{m,-}^\varepsilon)'(x)} \right)' dx.$$

The σ -derivatives are obtained by inserting the corresponding $\log x$ factors into the amplitudes. The endpoint certificate bounds these remainders by conservative integration-by-parts envelopes.

The endpoint sector certificate records three components:

first-cell plus $\zeta(2)$ cancellation,

Ξ_{end} coefficient correction,

and

post-IBP endpoint remainder.

The final safety-normalized endpoint constants are

$$C_{\text{end},0} = 71000100.000071000000,$$

and

$$C_{\text{end},1} = 281000100.000281000000.$$

Therefore

$$|\mathcal{R}_{\text{end}}(s)| \leq C_{\text{end},0} \frac{\log \tau}{\tau^{3/2}},$$

and

$$|\partial_\sigma \mathcal{R}_{\text{end}}(s)| \leq C_{\text{end},1} \frac{\log \tau}{\tau^{3/2}}.$$

5.3 The core sector

The core sector contains algebraic corrections left after the main quotient matching has been performed. It is marked as

side = mixed

because its subcomponents may have different normalizations. The explicit component certified in the final run is the quotient-side coefficient mismatch.

The quotient stationary coefficient ratio is

$$\rho_Q(s, \tau) = \frac{s-1}{s} \frac{C_A(s, \tau)}{C_B(s, \tau)}.$$

The corresponding J -packet ratio is

$$\rho_J(s, \tau) = -i \frac{s(s-2)}{(1-s)(s+1)} \left(\frac{2\pi}{\tau} \right)^{1-2\sigma} e^{2i\Theta(\tau)}.$$

The exact mismatch is

$$\Delta_\rho(s, \tau) = \rho_Q(s, \tau) - \rho_J(s, \tau).$$

A direct simplification gives

$$\Delta_\rho(s, \tau) = -i \left(\frac{2\pi}{\tau} \right)^{1-2\sigma} e^{2i\Theta(\tau)} \frac{s^2 - s + 1}{s(1-s)(s+1)}.$$

The corresponding core contribution is

$$G_\rho^{\text{core}}(s) = C_B(s, \tau) \Delta_\rho(s, \tau) \sum_{m \leq N(\tau)} m^{-s},$$

with

$$N(\tau) = \left\lfloor \frac{\tau}{4\pi} \right\rfloor.$$

This term is already in quotient normalization. It must not be multiplied by Π_Q again.

The derivative is

$$\begin{aligned} \partial_\sigma G_\rho^{\text{core}} &= (\partial_\sigma C_B) \Delta_\rho \sum_{m \leq N} m^{-s} \\ &\quad + C_B (\partial_\sigma \Delta_\rho) \sum_{m \leq N} m^{-s} \\ &\quad + C_B \Delta_\rho \partial_\sigma \sum_{m \leq N} m^{-s}. \end{aligned}$$

Here

$$\partial_\sigma C_B = \log \left(\frac{2\pi}{\tau} \right) C_B,$$

and

$$\partial_\sigma \sum_{m \leq N} m^{-s} = - \sum_{m \leq N} (\log m) m^{-s}.$$

Writing

$$Q(s) = \frac{s^2 - s + 1}{s(1-s)(s+1)},$$

one may compute

$$\partial_\sigma \Delta_\rho = -i \left(\frac{2\pi}{\tau} \right)^{1-2\sigma} e^{2i\Theta} \left[-2 \log \left(\frac{2\pi}{\tau} \right) Q(s) + Q'(s) \right].$$

The certificate bounds the Dirichlet polynomial and its derivative by elementary integral estimates. Since this component is Q -side, the core certificate records

projection = none

for this subcomponent.

The final safety-normalized core constants are

$$C_{\text{core},0} = 100.15143013245947061906516300167251169542246109728103520969123297128614893010207,$$

and

$$C_{\text{core},1} = 101.72541371865680722623693802471509402877474481545266923344493309421933909293843.$$

Thus

$$|\mathcal{R}_{\text{core}}(s)| \leq C_{\text{core},0} \frac{\log \tau}{\tau^{3/2}},$$

and

$$|\partial_\sigma \mathcal{R}_{\text{core}}(s)| \leq C_{\text{core},1} \frac{\log \tau}{\tau^{3/2}}.$$

Combining these three subsectors gives the band, endpoint, and core entries of the final certificate. Their constants are included in the final sums

$$C_0 = \sum_{\nu} C_{\nu,0}^{\text{safe}}, \quad C_1 = \sum_{\nu} C_{\nu,1}^{\text{safe}}.$$

The normalization ledger ensures that the band and core quotient-side pieces are not projected twice, while the endpoint sector is projected exactly once and includes the derivative of the projection.

Theorem 5.1 (Certified band, endpoint, and core sectors). *On*

$$\sigma \in [0.49, 0.51], \quad \tau \geq 10,$$

the three sectors satisfy

$$|\mathcal{R}_{\text{band}}(s)| \leq C_{\text{band},0} \frac{\log \tau}{\tau^{3/2}}, \quad |\partial_{\sigma} \mathcal{R}_{\text{band}}(s)| \leq C_{\text{band},1} \frac{\log \tau}{\tau^{3/2}},$$

with

$$C_{\text{band},0} = 5100100.000005100000, \quad C_{\text{band},1} = 9400100.000009400000;$$

$$|\mathcal{R}_{\text{end}}(s)| \leq C_{\text{end},0} \frac{\log \tau}{\tau^{3/2}}, \quad |\partial_{\sigma} \mathcal{R}_{\text{end}}(s)| \leq C_{\text{end},1} \frac{\log \tau}{\tau^{3/2}},$$

with

$$C_{\text{end},0} = 71000100.000071000000, \quad C_{\text{end},1} = 281000100.000281000000;$$

and

$$|\mathcal{R}_{\text{core}}(s)| \leq C_{\text{core},0} \frac{\log \tau}{\tau^{3/2}}, \quad |\partial_{\sigma} \mathcal{R}_{\text{core}}(s)| \leq C_{\text{core},1} \frac{\log \tau}{\tau^{3/2}},$$

with

$$C_{\text{core},0} = 100.15143013245947061906516300167251169542246109728103520969123297128614893010207,$$

and

$$C_{\text{core},1} = 101.72541371865680722623693802471509402877474481545266923344493309421933909293843.$$

Proof. The band sector is certified directly in quotient normalization by bounding the exact finite band residual after subtracting its endpoint main term. The endpoint sector is first bounded on the J -side and then projected by Π_Q , with the derivative term $(\partial_{\sigma} \Pi_Q) G_{\text{end}}^J$ included. The core sector records the explicit quotient-side coefficient mismatch and therefore applies no further projection to that subcomponent. Each sector certificate contains the value constant, the σ -derivative constant, a coverage hash, a normalization hash, and a proof hash. The displayed constants are the safety-normalized constants accepted by the final merger. $\square$

6 Far stationary and global nonstationary sector certificates

We now describe the two remaining analytic tail sectors:

$$\mathcal{R}_{\text{far}} \quad \text{and} \quad \mathcal{R}_{\text{nonstat}}.$$

Both sectors are naturally produced on the J -side and are therefore declared in the certificate with

$$\text{side} = \text{J_projected}.$$

Consequently, both sectors are projected into quotient normalization by

$$\Pi_Q(s) = \frac{i\tau}{(1-s)(s+1)}.$$

If

$$G_{\text{far}}^J(s) \quad \text{and} \quad G_{\text{nonstat}}^J(s)$$

denote the corresponding J -side residuals, then

$$\mathcal{R}_{\text{far}}(s) = \Pi_Q(s)G_{\text{far}}^J(s),$$

and

$$\mathcal{R}_{\text{nonstat}}(s) = \Pi_Q(s)G_{\text{nonstat}}^J(s).$$

The derivative certificates therefore include the projection derivative:

$$\partial_\sigma \mathcal{R}_{\text{far}}(s) = (\partial_\sigma \Pi_Q(s))G_{\text{far}}^J(s) + \Pi_Q(s)\partial_\sigma G_{\text{far}}^J(s),$$

and

$$\partial_\sigma \mathcal{R}_{\text{nonstat}}(s) = (\partial_\sigma \Pi_Q(s))G_{\text{nonstat}}^J(s) + \Pi_Q(s)\partial_\sigma G_{\text{nonstat}}^J(s),$$

where

$$\partial_\sigma \Pi_Q(s) = \Pi_Q(s) \left(\frac{1}{1-s} - \frac{1}{s+1} \right).$$

This projection derivative is part of the normalization contract and is recorded by the sector normalization hashes.

6.1 The far stationary sector

The far stationary sector accounts for the stationary-family residuals that remain after subtracting the leading stationary packets away from the boundary transition near $x = 2$. The relevant stationary families are

$$I_{m,+}^-(s) = \int_2^\infty x^{\sigma-3} e^{i(-2\pi m x + \tau \log x)} dx,$$

and

$$I_{m,-}^+(s) = \int_2^\infty x^{-\sigma-2} e^{i(2\pi m x - \tau \log x)} dx.$$

Both have stationary point

$$x_{m,*} = \frac{\tau}{2\pi m}.$$

This stationary point lies in the J -tail $x \geq 2$ precisely when

$$m \leq \frac{\tau}{4\pi}.$$

As in the boundary analysis, set

$$\alpha = \frac{\tau}{4\pi}, \quad w = \tau^{-1/2}, \quad r_m = w(m - \alpha).$$

The boundary transition corresponds to r_m bounded. The far stationary sector is separated from the boundary by imposing

$$r_m \leq -R_{\text{far}},$$

with the certified reference value

$$R_{\text{far}} = 4.$$

Equivalently, the far stationary mode set is

$$\mathcal{M}_{\text{far}}(\tau) = \left\{ m \in \mathbb{N} : m \leq \frac{\tau}{4\pi} - R_{\text{far}}\sqrt{\tau} \right\}.$$

This separation ensures that the stationary point is uniformly away from the endpoint $x = 2$. The endpoint-stationary transition window is not included in this sector; it belongs to the boundary sector.

The leading stationary phase is

$$\Phi_m(\tau) = \tau \log \left(\frac{\tau}{2\pi m} \right) - \tau.$$

For the plus-amplitude stationary family, the leading stationary packet is

$$\text{SP}_{m,+}(s) = (2\pi)^{5/2-\sigma} \tau^{\sigma-5/2} m^{2-\sigma} e^{i\Phi_m(\tau)-i\pi/4}.$$

For the minus-amplitude stationary family, the leading stationary packet is

$$\text{SP}_{m,-}(s) = (2\pi)^{\sigma+3/2} \tau^{-\sigma-3/2} m^{\sigma+1} e^{-i\Phi_m(\tau)+i\pi/4}.$$

The exact far stationary remainders are defined by

$$D_{m,+}^{\text{far}}(s) = I_{m,+}^-(s) - \text{SP}_{m,+}(s),$$

and

$$D_{m,-}^{\text{far}}(s) = I_{m,-}^+(s) - \text{SP}_{m,-}(s).$$

This is the object certified by the far sector: the exact integral minus the explicit leading stationary packet. No asymptotic symbol is part of the definition of the residual.

The J -side far residual is

$$G_{\text{far}}^J(s) = \sum_{m \in \mathcal{M}_{\text{far}}(\tau)} \frac{1}{4\pi^2 m^2} \left[c_+(s) D_{m,+}^{\text{far}}(s) + c_-(s) D_{m,-}^{\text{far}}(s) \right],$$

where

$$c_+(s) = s(s-2), \quad c_-(s) = (1-s)(s+1).$$

The derivative is

$$\begin{aligned} \partial_\sigma G_{\text{far}}^J(s) = \sum_{m \in \mathcal{M}_{\text{far}}(\tau)} \frac{1}{4\pi^2 m^2} & \left[c'_+(s) D_{m,+}^{\text{far}} + c_+(s) \partial_\sigma D_{m,+}^{\text{far}} \right. \\ & \left. + c'_-(s) D_{m,-}^{\text{far}} + c_-(s) \partial_\sigma D_{m,-}^{\text{far}} \right], \end{aligned}$$

with

$$c'_+(s) = 2s-2, \quad c'_-(s) = -2s.$$

The σ -derivatives of the exact stationary remainders are also exact. Since the phases are independent of σ ,

$$\partial_\sigma I_{m,+}^-(s) = \int_2^\infty (\log x) x^{\sigma-3} e^{i(-2\pi mx + \tau \log x)} dx,$$

and

$$\partial_\sigma I_{m,-}^+(s) = - \int_2^\infty (\log x) x^{-\sigma-2} e^{i(2\pi mx - \tau \log x)} dx.$$

The stationary packets satisfy

$$\partial_\sigma \text{SP}_{m,+}(s) = \log\left(\frac{\tau}{2\pi m}\right) \text{SP}_{m,+}(s),$$

and

$$\partial_\sigma \text{SP}_{m,-}(s) = -\log\left(\frac{\tau}{2\pi m}\right) \text{SP}_{m,-}(s).$$

Thus

$$\partial_\sigma D_{m,+}^{\text{far}} = \partial_\sigma I_{m,+}^- - \log\left(\frac{\tau}{2\pi m}\right) \text{SP}_{m,+},$$

and

$$\partial_\sigma D_{m,-}^{\text{far}} = \partial_\sigma I_{m,-}^+ + \log\left(\frac{\tau}{2\pi m}\right) \text{SP}_{m,-}.$$

For uniform certification, the far verifier uses scaled stationary variables. Let

$$u = \frac{x}{x_{m,*}}, \quad x_{m,*} = \frac{\tau}{2\pi m}.$$

Define

$$\psi(u) = \log u - u + 1.$$

Then

$$\psi(1) = 0, \quad \psi'(1) = 0, \quad \psi''(1) = -1.$$

The scaled stationary integrals are

$$\mathfrak{I}_+(\sigma, \tau, \beta) = \int_\beta^\infty u^{\sigma-3} e^{i\tau\psi(u)} du,$$

and

$$\mathfrak{I}_-(\sigma, \tau, \beta) = \int_\beta^\infty u^{-\sigma-2} e^{-i\tau\psi(u)} du,$$

where

$$\beta = \frac{4\pi m}{\tau}.$$

The leading scaled stationary models are

$$\mathfrak{S}_+(\tau) = e^{-i\pi/4} \sqrt{\frac{2\pi}{\tau}},$$

and

$$\mathfrak{S}_-(\tau) = e^{i\pi/4} \sqrt{\frac{2\pi}{\tau}}.$$

The scaled remainders are

$$\mathfrak{D}_+ = \mathfrak{I}_+ - \mathfrak{S}_+, \quad \mathfrak{D}_- = \mathfrak{I}_- - \mathfrak{S}_-.$$

The far-sector envelope certifies these scaled remainders uniformly in the far stationary range, including their σ -derivatives. The final certificate uses a deliberately conservative envelope. The far sector safe constants are

$$C_{\text{far},0} = 50000000100.050000000000,$$

and

$$C_{\text{far},1} = 100000000100.100000000000.$$

Therefore

$$|\mathcal{R}_{\text{far}}(s)| \leq C_{\text{far},0} \frac{\log \tau}{\tau^{3/2}},$$

and

$$|\partial_\sigma \mathcal{R}_{\text{far}}(s)| \leq C_{\text{far},1} \frac{\log \tau}{\tau^{3/2}}.$$

6.2 The global nonstationary sector

The global nonstationary sector contains the four-family tail pieces that are not assigned to the band, boundary, endpoint, far stationary, floor, or core sectors. The four tail families are

$$I_{m,+}^\varepsilon(s) = \int_2^\infty x^{\sigma-3} e^{i(\varepsilon 2\pi m x + \tau \log x)} dx,$$

and

$$I_{m,-}^\varepsilon(s) = \int_2^\infty x^{-\sigma-2} e^{i(\varepsilon 2\pi m x - \tau \log x)} dx,$$

with

$$\varepsilon \in \{+1, -1\}.$$

The phase derivatives are

$$(\Phi_{m,+}^\varepsilon)'(x) = \varepsilon 2\pi m + \frac{\tau}{x},$$

and

$$(\Phi_{m,-}^\varepsilon)'(x) = \varepsilon 2\pi m - \frac{\tau}{x}.$$

For $\tau > 0$, the families

$$I_{m,+}^+, \quad I_{m,-}^-$$

are always nonstationary. The families

$$I_{m,+}^-, \quad I_{m,-}^+$$

may have stationary points, but only when

$$m \leq \frac{\tau}{4\pi}.$$

The stationary pieces in the far range are assigned to $\mathcal{R}_{\text{far}}$, and the endpoint-stationary transition window is assigned to $\mathcal{R}_{\text{bd}}$. The remaining parts are assigned to the global nonstationary sector.

Let

$$N \subset \mathbb{N} \times \{+, -\} \times \{+, -\}$$

be the certified nonstationary index set after removing the already assigned band, boundary, endpoint, far, floor, and core pieces. For

$$\kappa \in \{+, -\}, \quad \varepsilon \in \{+, -\},$$

write

$$I_{m,\kappa}^\varepsilon(s)$$

for the corresponding oscillatory integral. The J -side nonstationary residual is

$$G_{\text{nonstat}}^J(s) = \frac{1}{4\pi^2} \sum_{(m,\kappa,\varepsilon) \in N} \frac{1}{m^2} c_\kappa(s) I_{m,\kappa}^\varepsilon(s),$$

where

$$c_+(s) = s(s-2), \quad c_-(s) = (1-s)(s+1).$$

Its derivative is

$$\partial_\sigma G_{\text{nonstat}}^J(s) = \frac{1}{4\pi^2} \sum_{(m,\kappa,\varepsilon) \in N} \frac{1}{m^2} [c'_\kappa(s) I_{m,\kappa}^\varepsilon(s) + c_\kappa(s) \partial_\sigma I_{m,\kappa}^\varepsilon(s)].$$

Since the phases do not depend on σ ,

$$\partial_\sigma I_{m,+}^\varepsilon(s) = \int_2^\infty (\log x) x^{\sigma-3} e^{i\Phi_{m,+}^\varepsilon(x)} dx,$$

and

$$\partial_\sigma I_{m,-}^\varepsilon(s) = - \int_2^\infty (\log x) x^{-\sigma-2} e^{i\Phi_{m,-}^\varepsilon(x)} dx.$$

The certification is by integration by parts on regions where the phase derivative is separated from zero. For a general phase derivative

$$p(x) = \Phi'(x),$$

define

$$\mathcal{L}_p A = \left(\frac{A}{p} \right)' = \frac{A'}{p} - \frac{Ap'}{p^2}.$$

Then

$$\int_a^\infty A(x) e^{i\Phi(x)} dx = - \frac{A(a) e^{i\Phi(a)}}{ip(a)} - \frac{1}{i} \int_a^\infty \mathcal{L}_p A(x) e^{i\Phi(x)} dx.$$

A second integration by parts gives the certified bound

$$\left| \int_a^\infty A e^{i\Phi} dx \right| \leq \frac{|A(a)|}{|p(a)|} + \frac{|(\mathcal{L}_p A)(a)|}{|p(a)|} + \int_a^\infty |\mathcal{L}_p^2 A(x)| dx.$$

The derivative certificate uses the same operator with

$$A \text{ replaced by } \partial_\sigma A.$$

This is valid because the phases are independent of σ .

The global nonstationary certificate records a conservative envelope for the always-nonstationary families, the potentially stationary families outside their stationary ranges, and a partition guard ensuring that no region already assigned to another sector is counted again. The final safety-normalized constants are

$$C_{\text{nonstat},0} = 20000000100.020000000000,$$

and

$$C_{\text{nonstat},1} = 40000000100.040000000000.$$

Thus

$$|\mathcal{R}_{\text{nonstat}}(s)| \leq C_{\text{nonstat},0} \frac{\log \tau}{\tau^{3/2}},$$

and

$$|\partial_\sigma \mathcal{R}_{\text{nonstat}}(s)| \leq C_{\text{nonstat},1} \frac{\log \tau}{\tau^{3/2}}.$$

The final repository

<https://github.com/vicblab/RH-inner-strip-certifier>

contains the scripts and tutorial-style instructions used to reproduce these sector certificates, including the one-command driver

`./run_all_certification.sh.`

The repository README records the intended strip

$$\sigma \in [0.49, 0.51],$$

the bracket parameters

$$u_0 = 0.01, \quad g_0 = 0.2,$$

the target height

$$T = 10^{30},$$

and the expected merged constants

$$C_{\text{even}} \approx 2.5567524018976995 \times 10^{11},$$

$$C_{\sigma} \approx 2.9182458524367624 \times 10^{11},$$

$$C_{\text{inner}} \approx 3.879837844533457 \times 10^{11}.$$

It also states that the full run creates

`INNER_STRIP_CERTIFICATE_SAFE.json`

and that the final JSON must contain

`status = inner_full_certificate`

and

`height_condition_holds = true.`

This repository is the public reproduction artifact for the certified constants quoted in this paper. [11]

We summarize the far and global nonstationary sectors as follows.

Theorem 6.1 (Certified far and global nonstationary sectors). *On*

$$\sigma \in [0.49, 0.51], \quad \tau \geq 10,$$

the far stationary residual satisfies

$$|\mathcal{R}_{\text{far}}(s)| \leq C_{\text{far},0} \frac{\log \tau}{\tau^{3/2}},$$

and

$$|\partial_{\sigma} \mathcal{R}_{\text{far}}(s)| \leq C_{\text{far},1} \frac{\log \tau}{\tau^{3/2}},$$

with

$$C_{\text{far},0} = 50000000100.050000000000,$$

and

$$C_{\text{far},1} = 100000000100.100000000000.$$

The global nonstationary residual satisfies

$$|\mathcal{R}_{\text{nonstat}}(s)| \leq C_{\text{nonstat},0} \frac{\log \tau}{\tau^{3/2}},$$

and

$$|\partial_{\sigma} \mathcal{R}_{\text{nonstat}}(s)| \leq C_{\text{nonstat},1} \frac{\log \tau}{\tau^{3/2}},$$

with

$$C_{\text{nonstat},0} = 20000000100.020000000000,$$

and

$$C_{\text{nonstat},1} = 40000000100.040000000000.$$

Both sectors are J-side projected and include the derivative of the quotient projection.

Proof. The far sector is the exact stationary-family residual after subtracting the leading stationary packets in the region $r_m \leq -R_{\text{far}}$. The stationary point is uniformly separated from the endpoint $x = 2$, and the scaled stationary variable reduces the certification to uniform bounds for the scaled remainders and their σ -derivatives. The global nonstationary sector consists of the remaining four-family tail pieces after removing the band, boundary, endpoint, far, floor, and core sectors. On this set the phase derivative is bounded away from zero, so repeated integration by parts gives uniform value and σ -derivative bounds. Both J -side residuals are then projected by Π_Q , and the derivative certificates include $(\partial_\sigma \Pi_Q)G^J$. The displayed constants are the safety-normalized constants recorded in the final sector JSON files and accepted by the final strict merger. $\square$

7 Final merge and height computation

We now assemble the seven residual-sector certificates and the finite-width bracket certificate into the final inner-strip certificate. The final merger is the strict bracket-aware verifier

`merge_inner_certificate_with_bracket_safe.py.`

It requires the seven residual-sector files

`band_sector.safe.json, bd_sector.safe.json, far_sector.safe.json, end_sector.safe.json,`

`nonstat_sector.safe.json, floor_sector.safe.json, core_sector.safe.json,`

and it also requires the finite-width bracket certificate

`inner_bracket_vector.safe.json.`

The final certificate is not allowed to be produced from the residual-sector certificates alone. This is important: the obstruction requires both the residual upper bound and the finite-width bracket lower bound.

The strict final merger checks the following conditions.

First, every residual-sector file must have

`status = proved.`

Second, every residual sector must declare one of the permitted normalization sides:

`Q, J_projected, zero, mixed.`

Third, every residual sector must contain both safety-normalized constants

`C_value_upper_safe`

and

`C_sigma_derivative_upper_safe.`

Fourth, the finite-width bracket certificate must have

`status = proved`

and must certify

`g0_bracket_vector_lower  $\geq$  0.2.`

Only after these conditions are satisfied does the final merger compute the residual vector constant and check the height inequality.

Let

$$C_{\nu,0}^{\text{safe}}$$

and

$$C_{\nu,1}^{\text{safe}}$$

denote the value-channel and σ -derivative-channel constants for a sector

$$\nu \in \{\text{band}, \text{bd}, \text{far}, \text{end}, \text{nonstat}, \text{floor}, \text{core}\}.$$

The final value-channel constant is

$$C_0 = \sum_{\nu} C_{\nu,0}^{\text{safe}}.$$

The final derivative-channel constant is

$$C_1 = \sum_{\nu} C_{\nu,1}^{\text{safe}}.$$

The two channels are then combined as a Euclidean vector bound:

$$C_{\text{inner}} = \sqrt{C_0^2 + C_1^2}.$$

In the final certificate, the seven sector safe constants are as follows.

For the band sector,

$$C_{\text{band},0}^{\text{safe}} = 5100100.000005100000,$$

and

$$C_{\text{band},1}^{\text{safe}} = 9400100.000009400000.$$

For the boundary sector,

$$C_{\text{bd},0}^{\text{safe}} = 185599139689.54844424576273899129572044225947553844933704222323532721371511756646,$$

and

$$C_{\text{bd},1}^{\text{safe}} = 151534184741.81053762160606677355438429846417653094438319187408516674070499261542.$$

For the far stationary sector,

$$C_{\text{far},0}^{\text{safe}} = 50000000100.050000000000,$$

and

$$C_{\text{far},1}^{\text{safe}} = 100000000100.100000000000.$$

For the endpoint sector,

$$C_{\text{end},0}^{\text{safe}} = 71000100.000071000000,$$

and

$$C_{\text{end},1}^{\text{safe}} = 281000100.000281000000.$$

For the global nonstationary sector,

$$C_{\text{nonstat},0}^{\text{safe}} = 20000000100.020000000000,$$

and

$$C_{\text{nonstat},1}^{\text{safe}} = 40000000100.040000000000.$$

For the floor sector,

$$C_{\text{floor},0}^{\text{safe}} = 0, \quad C_{\text{floor},1}^{\text{safe}} = 0.$$

For the core sector,

$$C_{\text{core},0}^{\text{safe}} = 100.15143013245947061906516300167251169542246109728103520969123297128614893010207,$$

and

$$C_{\text{core},1}^{\text{safe}} = 101.72541371865680722623693802471509402877474481545266923344493309421933909293843.$$

Summing the value-channel constants gives

$$\begin{aligned} C_0 &= C_{\text{band},0}^{\text{safe}} + C_{\text{bd},0}^{\text{safe}} + C_{\text{far},0}^{\text{safe}} + C_{\text{end},0}^{\text{safe}} \\ &\quad + C_{\text{nonstat},0}^{\text{safe}} + C_{\text{floor},0}^{\text{safe}} + C_{\text{core},0}^{\text{safe}} \\ &= 255675240189.76995047822220961036088344393198723387179813950427053690494808885261. \end{aligned}$$

Summing the derivative-channel constants gives

$$\begin{aligned} C_1 &= C_{\text{band},1}^{\text{safe}} + C_{\text{bd},1}^{\text{safe}} + C_{\text{far},1}^{\text{safe}} + C_{\text{end},1}^{\text{safe}} \\ &\quad + C_{\text{nonstat},1}^{\text{safe}} + C_{\text{floor},1}^{\text{safe}} + C_{\text{core},1}^{\text{safe}} \\ &= 291824585243.67624174026287399979132232317927055971912800732675440018563808683476. \end{aligned}$$

Therefore

$$C_{\text{inner}} = \sqrt{C_0^2 + C_1^2}$$

is

$$C_{\text{inner}} = 387983784453.34570013486357394399212036290592556640627403482826815918742177195616.$$

This value is recorded in the final JSON as

$$\mathbf{C_inner_vector_upper}.$$

The bracket certificate gives

$$g_0 = 0.2.$$

The final obstruction inequality is

$$\sqrt{T} > \frac{C_{\text{inner}}}{g_0} \log T.$$

Substituting the certified values gives

$$\frac{C_{\text{inner}}}{g_0} = 1939918922266.7285006743178697199606018145296278320313701741413407959371088597808.$$

Thus the inequality to be checked is

$$\sqrt{T} > 1939918922266.7285006743178697199606018145296278320313701741413407959371088597808 \log T.$$

The final certificate checks this inequality at

$$T = 10^{30}.$$

At this height,

$$\sqrt{T} = 10^{15},$$

while

$$\frac{C_{\text{inner}}}{g_0} \log T = 1939918922266.7285006743178697199606018145296278320313701741413407959371088597808.3$$

The strict merger records

$$\text{height_condition_holds} = \text{true}.$$

Thus the certified height condition holds at the shared target height

$$T = 10^{30}.$$

The height threshold can also be solved directly. Setting

$$\sqrt{T} = \frac{C_{\text{inner}}}{g_0} \log T$$

and solving numerically gives

$$T_{\min} \approx 1.5867647318922528 \times 10^{28}.$$

Equivalently,

$$\log_{10} T_{\min} \approx 28.2005125390955.$$

Therefore the clean power-of-ten height statement implied by the certificate is

$$T \geq 10^{29}.$$

The repository provides the script

$$\text{tools/height_threshold.py}$$

to reproduce this computation. The public repository

$$\text{https://github.com/vicblab/RH-inner-strip-certifier}$$

also records the intended target strip, the expected merged constants, the one-command reproduction script, and the final JSON condition

$$\text{status} = \text{inner_full_certificate}, \quad \text{height_condition_holds} = \text{true}.$$

The README states that running

$$\text{./run_all_certification.sh}$$

creates all sector certificates in

$$\text{certs_inner/}$$

and then creates

$$\text{INNER_STRIP_CERTIFICATE_SAFE.json}.$$

This is the public reproducibility artifact for the final constants used in this paper. [11] We now state the final certified merge theorem.

Theorem 7.1 (Strict final inner-strip certificate). *The strict bracket-aware final merger produces the certificate*

`INNER_STRIP_CERTIFICATE_SAFE.json`

with

`status = inner_full_certificate`

and

`height_condition_holds = true.`

The certificate records

$C_0 = 255675240189.76995047822220961036088344393198723387179813950427053690494808885261,$

$C_1 = 291824585243.67624174026287399979132232317927055971912800732675440018563808683476,$

and

$C_{\text{inner}} = 387983784453.34570013486357394399212036290592556640627403482826815918742177195616.$

Together with

$$g_0 = 0.2,$$

the height inequality

$$\sqrt{T} > \frac{C_{\text{inner}}}{g_0} \log T$$

holds at

$$T = 10^{30}.$$

Moreover, the numerical threshold is

$$T_{\min} \approx 1.5867647318922528 \times 10^{28},$$

so the clean certified height statement is

$$T \geq 10^{29}.$$

Proof. The final merger first verifies that all seven residual sectors are present and have status **proved**. It then verifies that the finite-width bracket certificate is present and has status **proved**, with

$$g_0_bracket_vector_lower \geq 0.2.$$

It sums the seven value-channel constants to obtain C_0 , sums the seven σ -derivative-channel constants to obtain C_1 , and computes

$$C_{\text{inner}} = \sqrt{C_0^2 + C_1^2}.$$

Finally, it evaluates

$$\sqrt{T} > \frac{C_{\text{inner}}}{g_0} \log T$$

at

$$T = 10^{30}.$$

The computed Boolean field

`height_condition_holds`

is true. Therefore the strict final certificate is issued. □

Combining the strict final certificate with the structural quotient obstruction gives the desired exclusion statement.

Corollary 7.2 (Certified central inner-strip exclusion above 10^{29}). *Assume the structural quotient identity and two-channel obstruction theorem proved in the analytic companion paper. Then the Riemann zeta function has no off-critical nontrivial zeros in the central inner strip*

$$0 < |\Re s - \tfrac{1}{2}| \leq 0.01$$

for

$$|\Im s| \geq 10^{29}.$$

In particular, the exclusion holds at the target height

$$|\Im s| \geq 10^{30}.$$

Proof. At a nontrivial zero, the structural identity gives

$$\mathcal{N}_Q(s) = 0.$$

The certified expansion gives

$$\mathcal{N}_Q(s) = \frac{i}{\tau} \mathfrak{B}_{\text{eff}}(\sigma, \tau) + \mathcal{R}_{\text{total}}(s).$$

The finite-width bracket certificate gives

$$\|\mathfrak{B}(u, \tau)\| \geq g_0 = 0.2$$

for

$$0 \leq u \leq 0.01.$$

The residual certificate gives

$$\|\mathcal{R}_{\text{total}}(s)\| \leq C_{\text{inner}} \frac{\log \tau}{\tau^{3/2}}.$$

If

$$\sqrt{\tau} > \frac{C_{\text{inner}}}{g_0} \log \tau,$$

then the residual is too small to cancel the bracket term. Hence

$$\mathcal{N}_Q(s) \neq 0,$$

contradicting the zero assumption. The certified height computation shows that this inequality holds for

$$\tau \geq 10^{29}.$$

Thus there are no off-critical zeros in the stated inner strip above that height. $\square$

8 Global assembly with the outer-strip certificate

The certificate proved in this paper is an inner-strip certificate. Its certified region is

$$\Omega_{\text{in}} = \{s = \sigma + i\tau : 0 < |\sigma - \tfrac{1}{2}| \leq 0.01, |\tau| \geq 10^{29}\}.$$

It does not by itself certify the full critical strip. To obtain a global high-height zero exclusion away from the endpoints $\sigma = 0$ and $\sigma = 1$, one must combine it with the outer-strip certificate.

Let $\delta_{\text{out}} > 0$ denote the endpoint margin certified by the outer-strip argument. Assume that the outer-strip companion theorem [9] proves that there are no off-critical zeros in

$$\Omega_{\text{out}} = \{s = \sigma + i\tau : 0.01 \leq \sigma \leq 0.49, |\tau| \geq 10^{30}\} \cup \{s = \sigma + i\tau : 0.51 \leq \sigma \leq 0.99, |\tau| \geq 10^{30}\}.$$

Assume also that the outer-strip theorem and the present inner-strip theorem use compatible quotient-normalization conventions and compatible height normalizations.

The present paper proves the central part

$$0.49 \leq \sigma \leq 0.51, \quad \sigma \neq \tfrac{1}{2}, \quad |\tau| \geq 10^{29}.$$

Since

$$10^{30} \geq 10^{29},$$

the inner-strip certificate is valid throughout the central strip at the outer-strip target height 10^{30} . Therefore, after combining the outer and inner certificates, the covered high-height region is

$$0.01 \leq \sigma \leq 1 - 0.01, \quad \sigma \neq \tfrac{1}{2}, \quad |\tau| \geq 10^{30}.$$

Theorem 8.1 (Conditional global high-height assembly). *Assume the structural quotient-obstruction theorem, the finite-width inner-strip certificate proved in this paper, and the outer-strip certificate [10] on*

$$[0.01, 0.49] \cup [0.51, 0.99]$$

at height 10^{30} . Then there are no off-critical nontrivial zeros of $\zeta(s)$ in

$$0.01 \leq \Re s \leq 1 - 0.01, \quad \Re s \neq \tfrac{1}{2}, \quad |\Im s| \geq 10^{30}.$$

Proof. The outer-strip certificate excludes off-critical zeros for

$$0.01 \leq \sigma \leq 0.49 \quad \text{and} \quad 0.51 \leq \sigma \leq 0.99$$

at height 10^{30} . The present inner-strip certificate excludes off-critical zeros for

$$0.49 \leq \sigma \leq 0.51, \quad \sigma \neq \tfrac{1}{2},$$

already for height 10^{29} , and hence also for height 10^{30} . The union of these certified regions is

$$0.01 \leq \sigma \leq 0.99, \quad \sigma \neq \tfrac{1}{2}.$$

Thus no off-critical zero can occur in the stated region at height $|\tau| \geq 10^{30}$. $\square$

A Reproducibility and audit protocol

This appendix records the exact reproducibility protocol for the inner-strip certificate. The public repository for the certificate is

`https://github.com/vicblab/RH-inner-strip-certifier`.

The repository README states that the certifier reproduces the inner-strip residual constants for

$$\sigma \in [0.49, 0.51], \quad u_0 = 0.01, \quad g_0 = 0.2, \quad T = 10^{30},$$

and that the final certificate is the file

`INNER_STRIP_CERTIFICATE_SAFE.json`.

It also states that the expected merged constants are approximately

$$C_{\text{even}} \approx 2.5567524018976995 \times 10^{11},$$

$$C_{\sigma} \approx 2.9182458524367624 \times 10^{11},$$

and

$$C_{\text{inner}} \approx 3.879837844533457 \times 10^{11}.$$

The repository provides a one-command reproduction script

`./run_all_certification.sh`,

which generates all sector certificates in

`certs_inner/`

and then creates

`INNER_STRIP_CERTIFICATE_SAFE.json`.

The README further specifies that the final JSON must contain

`status = inner_full_certificate`

and

`height_condition_holds = true`.

These are the public reproduction conditions for the certificate. [11]

The repository is intended to be tested from a clean Python environment. A typical run begins with

`python -m venv .venv`,

followed by

`source .venv/bin/activate`,

`python -m pip install -upgrade pip`,

and

`python -m pip install python-flint`.

After installing the dependency, the full certificate is reproduced by running

`./run_all_certification.sh`.

This command creates the seven residual-sector certificates, the finite-width bracket certificate, and the strict final merged certificate.

The complete list of final sector certificates is

```
certs_inner/band_sector.safe.json,
certs_inner/bd_sector.safe.json,
certs_inner/far_sector.safe.json,
certs_inner/end_sector.safe.json,
certs_inner/nonstat_sector.safe.json,
certs_inner/floor_sector.safe.json,
```

and

```
certs_inner/core_sector.safe.json.
```

The final strict merger also requires

```
certs_inner/inner_bracket_vector.safe.json.
```

Thus the minimum complete certificate directory must contain eight files: seven residual-sector JSON files and one bracket JSON file.

The final theorem-level merger is

```
scripts/merge_inner_certificate_with_bracket_safe.py.
```

It is run as

```
python scripts/merge_inner_certificate_with_bracket_safe.py
-inputs 'certs_inner/*_sector.safe.json'
-bracket certs_inner/inner_bracket_vector.safe.json
-height 1e30
-g0 0.2
-output INNER_STRIP_CERTIFICATE_SAFE.json.
```

This strict merger checks both the residual certificates and the finite-width bracket certificate. It refuses to produce a theorem-level certificate if the bracket certificate is missing. This is essential because the residual constants alone do not prove the obstruction; the finite-width bracket lower bound is also required.

The final file

```
INNER_STRIP_CERTIFICATE_SAFE.json
```

must contain

```
status = inner_full_certificate,
height_condition_holds = true,
u0 = 0.01,
```

and

```
g0_bracket_vector_lower = 0.2.
```

It must also contain a

```
bracket_certificate
```

field recording the file and proof hash of

`certs_inner/inner_bracket_vector.safe.json.`

Finally, it must contain seven sector entries under the field

`sectors.`

The seven sector names must be exactly

`band, bd, far, end, nonstat, floor, core.`

B Sector audit checklist

Each residual-sector JSON is checked according to the same basic schema. The field

`status`

must equal

`proved.`

The field

`side`

must be one of

`Q, J_projected, zero, mixed.`

The fields

`C_value_upper_safe`

and

`C_sigma_derivative_upper_safe`

must be present and nonnegative. The fields

`coverage_hash, normalization_hash, proof_hash`

must also be present.

The permitted sector sides are fixed by the normalization ledger:

`band` $\mapsto$ `Q`,

`bd` $\mapsto$ `J_projected`,

`far` $\mapsto$ `J_projected`,

`end` $\mapsto$ `J_projected`,

`nonstat` $\mapsto$ `J_projected`,

`floor` $\mapsto$ `zero`,

and

`core` $\mapsto$ `mixed`.

The band sector is already in quotient normalization and is not projected again. The boundary, far, endpoint, and global nonstationary sectors are J -side sectors and are projected by Π_Q . Their derivative certificates include the term

$$(\partial_\sigma \Pi_Q) G^J.$$

The floor sector is identically zero. The core sector is mixed because the explicit coefficient mismatch is quotient-side, while any later algebraic subcomponents must declare their own normalization.

The boundary sector has its own internal audit. The file

`bd_sector.safe.json`

is assembled from the following hooks:

finite compact boundary,
midrange compact boundary bridge,
high- τ compact Taylor boundary,
compact-complement tail,
compact-complement nonstationary transition,

and

compact-complement stationary overlap.

The boundary sector is accepted only after these hooks are merged into the single boundary-sector certificate. The transition hooks record complementary box counts: phase-separated boxes are assigned to the nonstationary hook, and boxes whose derivative interval intersects the stationary threshold are assigned to the stationary-overlap hook. This prevents the compact-complement transition region from being silently dropped.

The bracket certificate is checked separately. It must record

`status = proved,`
`sector = inner_bracket_vector,`
`u0 = 0.01,`

and

`g0_bracket_vector_lower = 0.2.`

The strict final merger does not accept the earlier critical-line-only bracket record. It requires a finite-width bracket certificate.

C Expected final constants

The final JSON produced in the certified run records

$C_0 = 255675240189.76995047822220961036088344393198723387179813950427053690494808885261,$

$C_1 = 291824585243.67624174026287399979132232317927055971912800732675440018563808683476,$

and

$C_{\text{inner}} = 387983784453.34570013486357394399212036290592556640627403482826815918742177195616.$

The value

C_{inner}

is the Euclidean combination

$$C_{\text{inner}} = \sqrt{C_0^2 + C_1^2}.$$

The final bracket lower bound is

$$g_0 = 0.2.$$

Thus

$$\frac{C_{\text{inner}}}{g_0} = 1939918922266.7285006743178697199606018145296278320313701741413407959371088597808.$$

The final certificate checks the height inequality

$$\sqrt{T} > \frac{C_{\text{inner}}}{g_0} \log T$$

at

$$T = 10^{30}.$$

It records

$$\text{height_condition_holds} = \text{true}.$$

Solving the equality

$$\sqrt{T} = \frac{C_{\text{inner}}}{g_0} \log T$$

gives

$$T_{\min} \approx 1.5867647318922528 \times 10^{28}.$$

Therefore the clean power-of-ten certified height is

$$T \geq 10^{29}.$$

The repository contains the utility

$$\text{tools/height_threshold.py}$$

to reproduce this calculation.

D Hash and dependency policy

Every certificate JSON contains a proof hash. The proof hash is not intended to replace mathematical verification; rather, it prevents accidental transcription or substitution of JSON files during the merge. The final JSON records the proof hash of each of the seven sector certificates and the proof hash of the bracket certificate. It also records coverage hashes and normalization hashes. These serve as a dependency ledger.

The coverage hash records which part of the decomposition is being certified. For example, the boundary coverage depends on the compact finite range, midrange bridge, high- τ Taylor range, compact-complement tail, nonstationary transition, and stationary-overlap transition. The normalization hash records whether the sector is Q -side, J -side projected, mixed, or zero. In particular, it records whether the quotient projection and its σ -derivative are included.

A verifier or referee should treat the final JSON as valid only if all of the following hold:

$$\text{status} = \text{inner_full_certificate},$$

$$\text{height_condition_holds} = \text{true},$$

all seven residual sectors are present,

$$\text{bracket_certificate}$$

is present, and the bracket certificate has

$$\text{status} = \text{proved}.$$

This is the minimal non-debatable audit condition for the certificate.

E Repository reproduction commands

The complete reproduction procedure is as follows. First, clone the public repository:

```
git clone https://github.com/vicblab/RH-inner-strip-certifier.
```

Then enter the repository and create a fresh environment:

```
cd RH-inner-strip-certifier,  
  
python -m venv .venv,  
  
source .venv/bin/activate,  
  
python -m pip install -upgrade pip,  
  
python -m pip install python-flint.
```

The full certificate is reproduced by

```
./run_all_certification.sh.
```

The final output file is

```
INNER_STRIP_CERTIFICATE_SAFE.json.
```

The height threshold is reproduced by

```
python tools/height_threshold.py.
```

The repository README gives the same tutorial-level instructions and records the expected final constants and final JSON status fields. [11]

The certification documented in this paper should therefore be read as a reproducible numerical and interval-arithmetic certificate for the structural quotient obstruction interface. The analytic proof of that interface is external to this certificate paper; the present document verifies the constants, normalization ledger, bracket lower bound, and height computation used by the inner-strip obstruction.

F Bibliographic references

The following references are used for the analytic and computational context of the certification. The repository reference is the public reproduction artifact for the certificate. The remaining references are standard references for the Riemann zeta function, asymptotic analysis, interval arithmetic, and Arb-style ball arithmetic.

References

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- [9] V. Blanco Bataller, *Certified Outer-Strip Quotient Obstruction for the Riemann Zeta Function*, companion certification manuscript.
- [10] V. Blanco Bataller, *RH-outer-strip-certifier*, certifier repository for the outer-strip residual constants and high-height outer-strip exclusion.
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The repository contains the scripts, sector certificates, reproduction pipeline, and height-threshold utility for the certified inner-strip constants. The README records the target strip, bracket parameters, expected constants, one-command reproduction command, and final JSON status requirements.